NLP homework 4

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(Dated: Mar 26, 2026)

The source code of the project can be viewed on github: https://github.com/randnull/nlp

I. INTRODUCTION

This work aims to research an embedding-based classifier and compare different approaches to embeddings: pretrained and locally trained.

II. DATASET

The *ag news dataset* [1] was selected as main dataset for this work. It contains 127.500 samples of news titles and descriptions with 4 labels:

- World
- Sports
- Business
- Sci/Tech

III. MODELS DESCRIPTION

This work presents several models for comparison:

- Word2vec model trained on full train dataset samples (120.000 samples)
- Pretrained Word2vec model - *word2vec-google-news-300* [2] (3.5 GB of vectors)
- Embedding as trainable matrix (nn.embedding)

IV. DATA PREPROCESSING & PIPELINES

We have several models types and due to that we need to preprocess our dataset to process it to model.

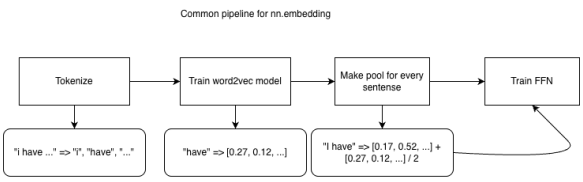

FIG. 1: common pipeline for w2v

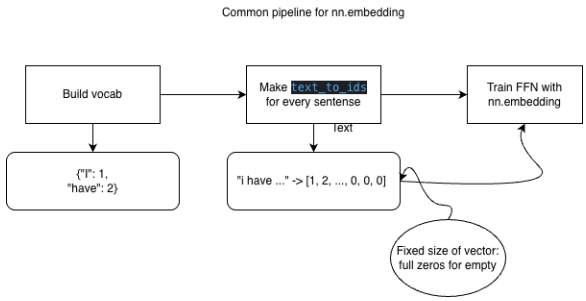

FIG. 2: common pipeline for nn.embedding

The pipeline for Word2Vec is shown in figure: 1: before feeding the input into the neural network we tokenize all sentenses, train word2vec models and then pool (mean) all words in sentense for every sentense.

The pipeline for nn.emdedding in figure: 2: we create a vocabulary of all words and represent each sentence as a list of word IDs.

V. FEEDFORWARD NEURAL NETWORK

In this work we compare not only different size of word2vec model's train datasets, but also different approaches to create embedding. It means that we need 2 different FNN architectures.

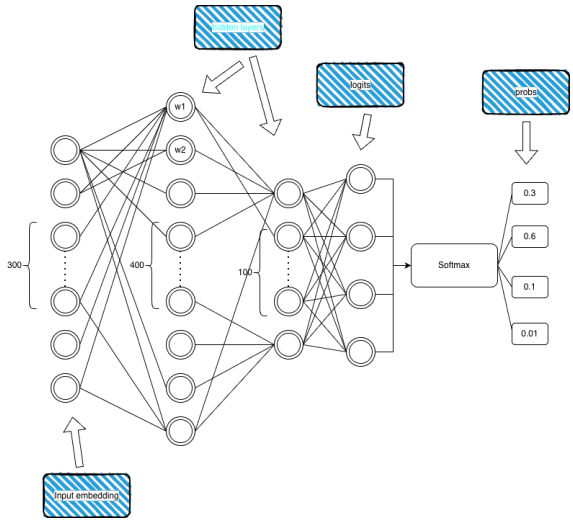

FIG. 3: FFN

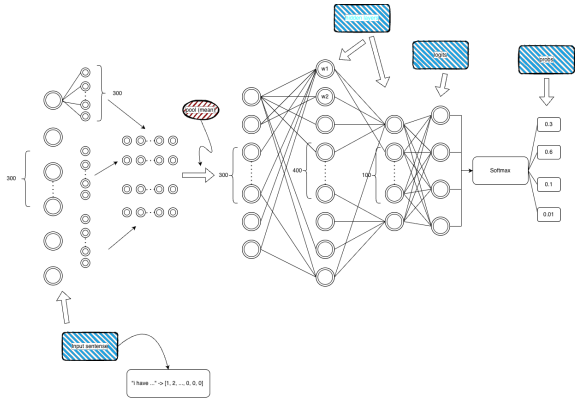

FIG. 4: FFN with embedding

First 3 architecture for embeddings from word2vec model has fixed input (embeddings), 2 hidden layes and output (softmax/argmax). Pool process (prepare sentences as embeddings) are in word2vec model.

Second 4 architecture larger than previous due to has additional layers to create embeddings of sentences. Fistly, we have only vector with number's of each word in sentense in vocabulary. Than we pool them to one vector (mean) and than repeat 3 architecture.

VI. METRICS & SCORES

To compare different models we will use several metrics:

- F1-score
- Accuracy
- Precision - macro
- Recall - macro

Table I provide resutls of all models by these metrics:

TABLE I: Comparison of models

Model	F1-score	Accuracy	Precision (m)	Recall (m)
w2v-full	0.9034	0.9034	0.9036	0.9034
w2v-pretrained	**0.9161**	**0.9162**	**0.9165**	**0.9162**
embedding	0.9023	0.9024	0.9022	0.9024

VII. RESULT

Pretrained word2vec model from google-news dataset gives best metric accross all models.

Overall, the results demonstrate that embeddings trained on large-scale external data provide the most informative semantic representations for the classification task. At the same time, locally trained Word2Vec models remain effective.

In contrast, embeddings learned only as part of the classification model are less reusable, since they are optimized specifically for the current task and cannot be directly transferred to other downstream applications.

[1] Huggingface, *ag news*, https://huggingface.co/datasets/sh0416/ag_news (accessed: 22 March 2026).
[2] Huggingface, *word2vec-google-news-300*, https://huggingface.co/fse/word2vec-google-news-300 (accessed: 22 March 2026).